



Procurement Management System(PMS)	Software Requirement Specification
Version ID Date: 01/02/2024 Document ID: SWD/01 Version ID: 1.0	

Revision History

Doc. Ver.	Date	Author	Reviewer	Description of Revision
1.0	01/02/2024	Ashiqur Rahman Sami	BJIT Academy	Initial Version

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1. Introduction

1.1 Purpose

The purpose of this document is to outline the requirements for the development of a system that facilitates **supplier registration**, **RFP (Request for Purchase) creation**, and **quotation management**. The system aims to streamline supplier account creation, RFP publishing, and the evaluation process, while offering insightful reporting and dashboard features for better decision-making.

1.2 Scope

The system will be developed as part of the **supply chain management** and will be used by **suppliers**, **reviewers**, and **approvers** to manage the supplier registration process, RFP creation, and quotation submission. The system will be built using **Odoo** and will include features such as email verification, supplier registration, RFP management, reporting, and a dynamic dashboard.

1.3 Intended Stakeholder

- **Suppliers:** External users who register on the portal, submit quotations, and manage their profiles.
- **Reviewers:** Internal users who create RFPs, review supplier registrations, and evaluate quotations.
- **Approvers:** Internal users who approve or reject supplier registrations and RFPs.

1.4 References

- Odoo Documentation: [Odoo Coding Guidelines](#)

1.5 Definitions, Acronyms, and Abbreviations

Term/Acronym	Definition
RFP	Request for Purchase
RFQ OTP	Request for Quotation
OTP	One-Time Password
OWL	Odoo Web Library
TIN	Tax Identification Number
IBAN	International Bank Account Number
SWIFT	Society for Worldwide Interbank Financial Telecommunication

2. Overall Description

2.1 Overview

The system will be integrated into the existing Odoo framework and will provide a seamless experience for suppliers, reviewers, and approvers. The system will include a **supplier portal** for registration and quotation submission, and a **backend interface** for reviewers and approvers to manage RFPs and supplier accounts.

2.2 Technical platform

- **Development Tools:** Odoo, Python, PostgreSQL
- **Web Server:** Nginx
- **Containerization:** Docker
- **Frontend Framework:** OWL (Odoo Web Library)
- **Operating System:** Linux

3. Functional Requirements

3.1 Overview

The system will include the following core features:

Serial No	Main Features	Description
1	Email Sign-up and OTP Validation	Allows users to register using email and OTP verification.
2	Supplier Registration Form Submission	Provides a form for suppliers to register their details.
3	Two-Step Verification Process	Ensures secure supplier account creation through a two-step review process.
4	RFP Creation, Publication & Quotation Submission	Allow reviewers to create RFPs and suppliers to submit quotations.
5	Report Preparation	Generates reports in HTML and Excel formats.
6	Dashboard Preparation	Provides a dynamic dashboard for data visualization
7	Deliverables Preparation	Ensure all deliverables are prepared and documented.

3.1.1. Feature/Function 1: Email Verification with OTP Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
FR-1.1	The system allow users to enter an email address for registration.	Email input field must be present.	None	TC-001
FR-1.2	The system shall verify if the email is already registered or blacklisted.	System must check email against existing records.	None	TC-002
FR-1.3	The system shall send an OTP to the email if it passes validation checks.	OTP must be sent to the email.	None	TC-003
FR-1.4	The system shall validate the OTP within a time limit.	OTP must be validated within 5 minutes.	None	TC-004

3.1.2. Feature/Function 2: Supplier Registration Form Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
FR-2.1	The system shall provide a registration form divided into 5 sections.	Form must include Company Information, Financial Information, Previous Client References, Certification, and Document Submission.	None	TC-005

FR-2.2	Include Company Information	Mandatory Fields: Company name, address and company type. Form should also include company logo, primary contact, finance contact, and authorized contact Optional Fields: Trade license number, commencement date, expiry date and TIN number	Trade license number must be alphanumeric, 8-20 characters Commence date must be a date before today Expiry Date must be a future date if provided and alert if expired. Tax Identification Number (TIN) must be a 16-digit numeric string	TC-006
FR-2.3	Include Bank Information Fields	Mandatory Fields: Bank Name, Bank Address, IBAN, Optional Fields: SWIFT Code, Account Name	None	TC-007
FR-2.4	Include Previous Client References	Fields: Name, Email, Phone, Address	Up to 5 references must be provided. If any Email, Phone and Address is provided then name is mandatory.	TC-008
FR-2.5	Include Certification	Fields: Certification, Certificate Number, Certifying Body, Award Date and Expiry Date	Expired certification cannot be added	TC-009

FR-2.6	Document Submission	• Trade License / Business Registration • Certificate of Incorporation • Certificate of Good Standing • Establishment Card • VAT / Tax Certificate • Memorandum of Association / Power of Attorney / Legal Instrument authorizing • Authorized Person • Identification Document for Authorized Person / Signatory • Bank Letter indicating Bank Account Information • Past 2 years of Audited Financial Statements • Other Certification / Accreditations (Specify any additional documents not listed above)	None	TC-010
FR-2.7	Declaration	Declaration statement and a check button	If checked, then form can be submitted	TC-011
FR-2.8	The system shall validate mandatory fields and optional fields with specific validation rules.	Mandatory fields must be validated before submission.	None	TC-012
FR-2.9	The system shall allow document uploads with a maximum file size of 1 MB.	File upload must be restricted to 1 MB.	None	TC-013

3.1.3. Feature/Function 3: Two-Step Verification Process Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
FR-3.1	The system shall allow reviewers to review and approve/reject supplier registrations.	Reviewers must be able to approve, reject, or blacklist suppliers.	None	TC-014
FR-3.2	The system shall notify suppliers via email upon successful account creation.	Email must be sent to the supplier with login credentials.	None	TC-015

3.1.4. Feature/Function 4: RFP Creation and Quotation Submission

Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
FR-4.1	The system shall allow reviewers to create and publish RFPs in portal	Reviewers must be able to create RFPs in draft and submit them for approval.	None	TC-016
FR-4.2	The system shall allow suppliers to submit quotations(RFQ) against RFPs.	Suppliers must be able to submit quotations via the portal.	None	TC-017
FR-4.3	The system shall notify reviewers upon quotation submission.	Email notification must be sent to reviewers.	None	TC-018
FR-4.4	The system shall allow reviewers to evaluate quotations and recommend suppliers.	Reviewers must be able to score and recommend quotations.	None	TC-019

FR-4.5	Quotation Scoring Process	Quotations scored based on Price, Delivery, Terms	None	TC-020
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Criteria	Weightage (%)	Description
Price Competitiveness	40%	Lower price scores higher
Delivery Time	20%	Faster delivery scores higher
Warranty & Support	15%	Longer warranty scores higher
Longer warranty scores higher	10%	More flexible payment terms score higher
Supplier Reputation	15%	Based on past performance

Fig 1: Scoring Criteria (FR-4.5)

3.1.5. Feature/Function 5: Report Preparation

Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
FR-5.1	The system shall generate RFP reports in HTML and Excel formats.	Reports must include supplier information, RFP details, and product-wise breakdowns.	None	TC-021
FR-5.2	The system shall include supplier information, RFP details, and product-wise breakdowns in the reports.	Reports must be generated based on supplier, start date, and end date.	None	TC-022

3.1.6. Feature/Function 6: Dashboard Preparation

Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
FR-6.1	The system shall provide a dynamic dashboard for visualizing RFP data.	Dashboard must include metrics such as approved RFQs, total amount, and product-wise breakdowns.	None	TC-023
FR-6.2	The system shall allow users to select a supplier and specify a date range for metrics.	Dashboard must update based on selected supplier and date range.	None	TC-024

3.1.7. Feature/Function 7: Deliverables Preparation

Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
FR-7.1	The system shall ensure all deliverables are prepared and documented.	Deliverables must include Docker container, Nginx configuration, and proper documentation.	None	TC-025
FR-7.2	The system shall provide step-by-step installation and configuration guides.	Documentation must include dependency lists, installation steps, and troubleshooting guides.	None	TC-026

4. User Interface

UI No.	UI Name	Related Function Req ID	Description	Test case Identifier
UI-001	Supplier Registration Form	FR-2.1	Form for supplier registration.	TC-05
UI-002	RFP Creation Form	FR-4.1	Form for creating RFPs.	TC-016
UI-003	RFQ Creation Form	FR-4.2	Form for creating RFQ against RFP	TC-017
UI-004	Dashboard	FR-6.1	Dynamic dashboard for RFP data visualization.	TC-023

5. Non-Functional Requirements

5.1 Performance Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
NFR-1.1	The system shall be optimized for fast loading times.	Pages must load within 5 seconds.	None	TC-020
NFR-1.2	The system shall support real-time updates.	Updates must reflect immediately.	None	TC-021

5.2 Safety Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
NFR-2.1	The system shall ensure secure handling of sensitive data.	Data must be encrypted during transmission.	None	TC-022

5.3 Security Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
NFR-3.1	The system shall implement Odoo's existing user roles and permissions.	Users must only access data based on their roles.	None	TC-023

6. Design Constraints

- The system must be developed using **Odoo** and follow Odoo's coding guidelines.
- The system must be **Dockerized** and served behind **Nginx**.
- The system must be optimized for performance and responsive across various devices.

7. Software Quality Attributes

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
SQA-1	The system shall be thoroughly tested to ensure functionality, performance, and security.	All test cases must pass.	None	TC-024

8. Other Requirements

REQUIREMNT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
OR-1	The system shall be Dockerized.	The application must run within a Docker container.	None	TC-025